

RAJASEKHAR REDDY KOTHAPALLI

Software Engineer — Distributed Systems & Video Streaming

Overland Park, Kansas • +1 (913)-413-4174 • rxk40660@ucmo.edu • [GitHub](#) • [LinkedIn](#) • [Portfolio](#)

Summary

Software engineer with 2 years of professional development experience at Zoho writing production code for large-scale video streaming (VOD), edge delivery, and real-time communication (WebRTC) systems. Developed Java services and algorithms that cut video fetch and rendering times by 40%, built PoP edge-server software, and engineered adaptive bitrate streaming logic serving users across varying networks. M.S. in Computer Science (May 2026) with focus on distributed systems.

Work experience

Member Technical Staff, Zoho

Aug 2022 - Aug 2024

Java backend development for VOD, Livestreaming & WebRTC platforms

- **Video-On-Demand**

- Developed **advanced file-fetching modules in Java** for the Video-on-Demand platform, cutting video fetching and rendering times by **40%** and significantly improving streaming performance.
- Developed and integrated **Java-based data compression and optimization mechanisms (including GZIP compression)** within Video-on-Demand services, reducing network transmission overhead, improving **content delivery efficiency**, and **enhancing video playback** responsiveness.
- Engineered Java-based media delivery services utilizing **fMP4 (Fragmented MP4) architecture**, reducing **startup latency** and improving content delivery performance for **large-scale streaming applications**.
- Developed and optimized Java backend services for segment management and **adaptive bitrate streaming**, increasing throughput, scalability, and **playback quality** across varying network environments.
- Architected and implemented Java-based distributed **Point of Presence (PoP) services**, improving **content availability**, reducing **user-facing latency**, and enhancing system scalability **across geographically distributed regions**.
- Designed and implemented **Python-based load testing** suites using **Locust** to validate the **scalability** and performance of distributed backend services, **identifying bottlenecks** and supporting performance **optimization initiatives**.

- **WebRTC**

- Developed Java-based **MetaFile Manager** and **FileWatcher** components for distributed file systems, enabling real-time file discovery, monitoring, and efficient data retrieval across networked environments.
- Designed and implemented Java-based **file integrity validation mechanisms** using checksum and verification techniques, ensuring **secure file transfers** and preventing **data corruption/tampering** during transmission.
- Enhanced and optimized Java-based file synchronization and transfer services like **File-Copy** and **File-Sync** functionalities within **WebRTC**, improving **performance, reliability, and data synchronization accuracy** across distributed systems.

Projects

- **WhisperFlowApp:** Developed a real-time, offline speech-to-text desktop application using C# (.NET 8 WPF) and OpenAI's Whisper model (via Whisper.cpp) ensuring full data privacy.
- **Secure Chat Application:** Built a Java-based anonymous messaging system using WebSockets and encryption to enable private, registration-free communication.
- **Appliance-Warehouse:** An E-Commerce full-stack web application developed to simplify the process of buying and renting household appliances online.
- **Custom Python UNIX Shell:** Designed and implemented a UNIX-like shell in Python supporting core commands, piping, and I/O redirection with command auto-completion.
- **Iris-Pay:** Created a Python-based biometric payment prototype using iris recognition and Scikit-Learn for secure user authentication.
- **Battery Alarm:** Developed an Android app using Java that alerts users when the battery reaches full charge to prevent overcharging and extend battery life.
- **Block-chain:** TrustVault is a decentralized platform that leverages IPFS and the Sui blockchain to provide secure, transparent, and verifiable proof of ownership for digital content.

Technical Skills

- **Programming Languages :** Java, Python, C#, C, JavaScript
- **Web Technologies:** HTML5, CSS3, Node.js
- **Backend & API Development:** REST APIs, WebSockets, Microservices, gRPC, API Integration, Client-Server Architecture
- **Frameworks & Libraries:** Spring, Spring Boot, Spring AI, Flask, Django, .NET, WPF
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, CloudFront), Docker, Kubernetes, CI/CD
- **Databases & Storage:** MySQL, Oracle, MongoDB, Redis, SQL Query Optimization
- **Testing & Performance:** Unit Testing, Integration Testing, Load Testing (Locust), Performance Optimization
- **Version Control & Tools:** Git, GitHub, Azure DevOps, Postman, IntelliJ IDEA, Visual Studio
- **Software Engineering:** OOP, Data Structures & Algorithms, Design Patterns, SDLC, System Design, Distributed Systems, Multithreading, Debugging, Code Reviews

Education

Master of Science in Computer Science — University of Central Missouri Aug 2024 - May 2026

- Relevant Coursework: Advanced Web Applications & Services, Advanced Java Programming, Advanced Python Programming, Advanced C#/.NET Development, Advanced Operating Systems, Compiler Design, Advanced Algorithms and Database Systems, and Data Science.
- Engaged in **blockchain-based research** under faculty supervision as a Research Assistant.

Bachelor of Technology, Computer Science — Krishna University Aug 2019 - May 2022

- Relevant Coursework: Object-Oriented Programming (Java), Data Structures & Algorithms, Database Management Systems, Computer Networks, Software Engineering, Cloud Computing, Mobile Application Development, and Software Testing.

Additional information

Certifications :

- Secured Elite + Silver Grade in **Python For Data Science** conducted by National Programme on Technology Enhanced Learning India
- Secured A+ grade in Diploma in **Hardware & Networking exam** conducted by International Council of Education & Training (iiet).
- Attended a Workshop on **Ethical Hacking with Cyber Security** at IIT Hyderabad.